Extensions of Schur Numbers: Order-Invariance, Density Thresholds, and Multiplicative Variants

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Abstract

We report computational results in three directions extending classical Schur numbers. First, we study Schur numbers $S(G, k)$ for finite abelian groups G and find that $S(G, 1)$ and $S(G, 2)$ depend only on $|G|$, verified through order 20, while $S(G, 3)$ is *not* order-invariant: $S(\mathbb{Z}/9\mathbb{Z}, 3) = 8$ but $S(\mathbb{Z}/3\mathbb{Z}\times\mathbb{Z}/3\mathbb{Z}, 3) = 7$. Second, we introduce density Schur numbers $\mathrm{DS}(k, \alpha)$ and determine $\mathrm{DS}(2, \frac{1}{2}) = 5$ with a Lean 4 proof, identifying a phase transition in the parameter α. Third, we tabulate Rado numbers for several linear equations and compute multiplicative Schur numbers $\mathrm{MS}(k)$ on the domain $\{2, \ldots, N\}$, finding $\mathrm{MS}(1) = 3$ and $\mathrm{MS}(2) = 31$. All results are computational (exhaustive enumeration or SAT-based) and should be regarded as such until independent formal verification.

1 Schur numbers in finite abelian groups

1.1 Definitions

Let G be a finite abelian group written additively. A subset $C \subseteq G$ is *sum-free* if there are no $a, b, c \in C$ with $a + b = c$ (the case $a = b$ is permitted, so $\{0\}$ is never sum-free since $0 + 0 = 0$).

Definition 1.1. $S(G, k)$ is the maximum size of a subset $A \subseteq G$ admitting a partition $A = C_1 \sqcup \cdots \sqcup C_k$ into k sum-free sets.

Equivalently, $S(G, k) + 1$ is the minimum size of a subset of G that forces a monochromatic Schur triple in every k-coloring. When $G = \mathbb{Z}/n\mathbb{Z}$ and $k = 1$, the quantity $S(G, 1)$ is the maximum sum-free subset of $\mathbb{Z}/n\mathbb{Z}$.

For a single color ($k = 1$), Green and Ruzsa [1] established asymptotically tight bounds on the maximum sum-free subset of a general abelian group. For $\mathbb{Z}/p\mathbb{Z}$ with p an odd prime, the interval $\{x : p/3 < x < 2p/3\}$ is a maximum sum-free set of size $\lfloor p/3 \rfloor + O(1)$; we have verified this computationally for all primes $p \leq 19$.

1.2 Order-invariance

We say that $S(-, k)$ is *order-invariant* if $S(G, k)$ depends only on $|G|$ for all finite abelian groups G. Since any group isomorphism preserves sum-freeness, $S(G, k)$ is certainly an isomorphism invariant. The question is whether non-isomorphic groups of the same order can differ.

Observation 1.2 (Computational, verified through order 20)**.** $S(G, 1)$ and $S(G, 2)$ are order-invariant for finite abelian groups of order ≤ 20. This covers 30 non-isomorphic groups, including all 5 abelian groups of order 16 (which all satisfy $S(G, 1) = 8$ and $S(G, 2) = 12$).

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For $k = 1$, order-invariance may follow from the Green–Ruzsa theory [1], which determines $S(G, 1)$ in terms of $|G|$ and the group's structure constants. The $k = 2$ invariance appears to be a new observation.

Observation 1.3 (Computational)**.** $S(G, 3)$ is *not* order-invariant. The first failure occurs at order 9:

$$S(\mathbb{Z}/9\mathbb{Z},\, 3) = 8, \qquad S(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z},\, 3) = 7.$$

A second failure occurs at order 12: $S(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z},\, 3) = 11$ while $S(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z},\, 3) = 10$.

The underlying mechanism appears related to the group exponent: $\mathbb{Z}/9\mathbb{Z}$ has exponent 9 while $(\mathbb{Z}/3\mathbb{Z})^2$ has exponent 3, and the higher exponent allows more room for three independent sum-free color classes.

1.3 Tables

Table 1 records $S(\mathbb{Z}/n\mathbb{Z}, k)$ for small n and $k = 1, 2, 3$.

Table 1: Schur numbers $S(\mathbb{Z}/n\mathbb{Z}, k)$ for $n = 2, \ldots, 16$. All values computed by exhaustive enumeration.

n	$S(-,1)$	$S(-,2)$	$S(-,3)$	S_2/n
2	1	1	1	0.500
3	1	2	2	0.667
4	2	3	3	0.750
5	2	4	4	0.800
6	3	4	5	0.667
7	2	4	6	0.571
8	4	6	7	0.750
9	3	6	8	0.667
10	5	8	9	0.800
11	4	8	10	0.727
12	6	9		0.750
13	4	8		0.615
14	7	9		0.643
15	6	12		0.800
16	8	12		0.750

Blank entries in the $k = 3$ column indicate that the exhaustive computation (3^n colorings) was not run for those n due to time constraints. For $k = 3$, note that $S(\mathbb{Z}/n\mathbb{Z}, 3) = n - 1$ for all $n \leq 11$: the $n - 1$ nonzero elements of $\mathbb{Z}/n\mathbb{Z}$ can always be 3-colored with each color class sum-free, at least through $n = 11$.

Conjecture 1.4. *$S(G, 2)$ is order-invariant for all finite abelian groups.*

2 Density Schur numbers

Definition 2.1. For integers $k \geq 1$ and a real parameter $\alpha \in [0, 1]$, define the *density Schur number* $\mathrm{DS}(k, \alpha)$ as the minimum N such that every k-coloring of $\{1, \ldots, N\}$ has at least one color

class C_i that is *good*: either C_i contains a Schur triple ($a + b = c$ with $a, b, c \in C_i$ and $a, b, c \geq 1$) or $|C_i| > \alpha N$.

Remark 2.2 (Indexing caveat)**.** Definition 2.1 uses the 1-indexed domain $\{1, \ldots, N\}$, which is the natural setting for integer Schur triples. An alternative 0-indexed definition on $\{0, \ldots, N-1\}$ would give different values because $0 + 0 = 0$ is a "free" Schur triple that forces any color class containing 0 to be good by default. Our Lean 4 formalization [4] uses the 1-indexed convention (elements $\{1, \ldots, N\}$, represented internally as `Fin N` with the Schur condition $(a + 1) + (b + 1) = (c + 1)$).

Theorem 2.3 (Lean-verified)**.** $\mathrm{DS}(2, \frac{1}{2}) = 5$. *That is, for every 2-coloring of* $\{1, 2, 3, 4, 5\}$*, some color class either contains a Schur triple or has more than* $\frac{1}{2} \cdot 5 = 2.5$ *elements. The witness coloring on* $\{1, 2, 3, 4\}$ *that avoids this is* $C_0 = \{1, 4\}$*,* $C_1 = \{2, 3\}$*: both classes are sum-free and both have exactly* $2 \leq \lfloor 4/2 \rfloor$ *elements, so neither exceeds the density threshold.*

The proof is formalized in Lean 4 with zero sorry statements. The lower bound (exhibiting the witness coloring) and upper bound (pigeonhole: some class has $\geq 3 > 5/2$ elements) are both machine-checked.

Theorem 2.4 (Pigeonhole)**.** $\mathrm{DS}(k, 1/k) = S(k) + 1$*, where* $S(k)$ *is the classical Schur number.*

Proof. At $N = S(k) + 1$, every k-coloring of $\{1, \ldots, N\}$ forces a monochromatic Schur triple by the definition of $S(k)$, so every color class is good (the Schur-triple condition alone suffices). At $N = S(k)$, there exists a k-coloring with no monochromatic Schur triple. By pigeonhole, some class has $\geq \lceil N/k \rceil$ elements, but the density condition requires *strictly* more than N/k elements; when k divides N and the partition is balanced, this threshold is not exceeded. □

2.1 Phase transition for $k = 2$

A related quantity is the *density Ramsey number* $\mathrm{DS}'(k, \alpha)$: the minimum N such that every k-coloring of $\{1, \ldots, N\}$ has a color class C satisfying *both* $|C| \geq \alpha N$ *and* C contains a Schur triple (the conjunction, not the disjunction in Definition 2.1). Computational experiments reveal three regimes for $\mathrm{DS}'(2, \alpha)$:

1. **Regime 1** ($\alpha \leq 0.59$): $\mathrm{DS}'(2, \alpha) = 5$, matching $S(2) + 1$.
2. **Regime 2** ($0.60 \leq \alpha \leq 0.66$): $\mathrm{DS}'(2, \alpha) = 6$.
3. **Regime 3** ($\alpha \geq 0.67$): $\mathrm{DS}'(2, \alpha) > 12$.

The transition from Regime 1 to Regime 2 is sharp: at $\alpha = 0.60$, $N = 5$ has an avoiding coloring (one class can have 3 out of 5 elements, and $3 < 0.60 \times 5 = 3.0$, so the density condition is not triggered), but $N = 6$ does not. These values are determined by exhaustive enumeration over all 2^N colorings.

3 Rado numbers and multiplicative Schur

3.1 Rado numbers for linear equations

A Rado number $R(E, k)$ for an equation E is the minimum N such that every k-coloring of $\{1, \ldots, N\}$ has a monochromatic solution to E. We computed the following values using a SAT-based approach (CaDiCaL 1.5 solver with direct encoding).

The Schur values $R(a{+}b{=}c,\, k) = S(k) + 1$ are well-known ($S(1) = 1$, $S(2) = 4$, $S(3) = 13$, $S(4) = 44$). The van der Waerden values match $W(k; 3)$. The values $R(a{+}b{+}c{=}d,\, 2) = 11$ and $R(a{+}b{+}c{=}d,\, 3) = 43$ are verified by our SAT solver and are consistent with the known literature on Rado's theorem. A dash (—) indicates that the value was not computed (either trivial or exceeding our search bound).

Table 2: Rado numbers $R(E, k)$ for small equations and $k = 1, 2, 3$.

Equation E	$k = 1$	$k = 2$	$k = 3$
$a + b = c$ (Schur)	2	5	14
$a + b = 2c$ (van der Waerden)	—	9	27
$a + b + c = d$	—	11	43
$a + 2b = 3c$	—	13	—

3.2 Multiplicative Schur numbers

Definition 3.1. $\mathrm{MS}(k)$ is the minimum N such that every k-coloring of $\{2, 3, \ldots, N\}$ contains a monochromatic *multiplicative triple* $\{a, b, ab\}$ (with $a \leq b$).

The domain starts at 2 to exclude 1, which would trivially satisfy $1 \cdot a = a$ for all a.

Observation 3.2 (SAT-verified)**.** $\mathrm{MS}(1) = 3$ and $\mathrm{MS}(2) = 31$. Equivalently, $\{2, 3\}$ can be 1-colored avoiding multiplicative triples, but $\{2, 3, 4\}$ cannot (since $2 \cdot 2 = 4$). For $k = 2$, $\{2, \ldots, 31\}$ can be 2-colored, but $\{2, \ldots, 32\}$ cannot.

Additionally, SAT verification for $k = 3$ yields $\mathrm{MS}(3) = 16383 = 2^{14} - 1$ (with $\{2, \ldots, 16383\}$ admitting a valid 3-coloring but $\{2, \ldots, 16384\}$ not). This suggests the formula

$$\mathrm{MS}(k) = 2^{(3^k+1)/2} - 1, \tag{1}$$

which gives $\mathrm{MS}(1) = 2^2 - 1 = 3$, $\mathrm{MS}(2) = 2^5 - 1 = 31$, $\mathrm{MS}(3) = 2^{14} - 1 = 16383$. We emphasize that (1) is a *conjecture* extrapolated from three data points, not a theorem. The exponents $2, 5, 14$ satisfy the recurrence $f(k) = 3f(k - 1) - 1$ with $f(1) = 2$.

Multiplicative Schur triples have been studied recently by Mattos, Mergoni Cecchelli, and Parczyk [2], who investigate the equation $xy = z$ in the integers and obtain asymptotic density bounds. Our exact small-k values may complement their asymptotic results, but we have not verified whether the values $\mathrm{MS}(1)$ and $\mathrm{MS}(2)$ appear in prior work.

3.3 Mixed additive-multiplicative constraints

One can also forbid *both* additive Schur triples $\{a, b, a+b\}$ and multiplicative triples $\{a, b, ab\}$ simultaneously. We find that the resulting mixed number equals $S(k)+1$ for $k = 1, 2, 3$: the additive constraint completely dominates. This is consistent with the much larger values of $\mathrm{MS}(k)$ compared to $S(k) + 1$ (e.g., $\mathrm{MS}(2) = 31$ vs. $S(2) + 1 = 5$).

Acknowledgments

Computational analysis performed with AI assistance (Claude, Anthropic). All SAT-based results verified independently using CaDiCaL. The Lean 4 formalization of $\mathrm{DS}(2, \frac{1}{2}) = 5$ was machine-checked with zero sorry statements. Source code and Lean proofs are available at [4].

References

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